CircleNet with point-to-point attention

Figure 5-inspired layout · implemented mainline model · red frames = added modules

(a) Full network

Input
Q: 500 points
C: 2000 points

→ **KNN**
K = 16 / 128

→ **Feature**
extraction

→ **Two-token**
fusion

→ NEW **Point-to-point**
attention

→ **Boundary**
+ weight heads

→ NEW **Plane + soft**
circle fit

center + radius

(b) Feature extraction

Q [500,3] + C [2000,3]

Local patch
K = 16

Wide patch
K = 128

Conv1d ×3
max pool
128 channels

Conv1d ×3
max pool
128 channels

Two feature tokens / point

Shared across all 500 query points

(c) Feature fusion and point attention

Two-token Transformer · 1 layer, 4 heads
mean → F [500,128]

Cross-point module

NEW **Position MLP**
XYZ → 128; add to F

NEW **Morton order**
sort 500 points

NEW **Window attention**
32 points, 4 heads

NEW **Shifted windows**
shift = 16 points

NEW **Global attention · 500 points**

(d) Heads and circle fitting

Enriched point features [500,128]

Boundary
P [500,2]

Weight
W [500,1]

NEW **Plane normal**
n [3]

NEW **Soft fit weights**
$q = p^2 \cdot (0.05 + 0.95\ \mathrm{sigmoid}(w))$

NEW **Differentiable weighted circle fit**
Q + q + n → center c, radius r

One circle per positive patch

Added relative to the paper source implementation

Paper source: hard selection → PCA plane → weighted 2D least squares